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BATCH-1

ROBUST SPEECH EMOTION RECOGNITION USING CNN+LSTM BASED ON STOCHASTIC FRACTAL SEARCH OPTIMIZATION ALGORITHM

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ABSTRACT

This project addresses the challenge of limited data in speech emotion recognition by introducing a novel data augmentation algorithm. The proposed approach enriches the dataset by adding carefully designed noise fractions. Additionally, the paper presents an optimized deep learning model with hyperparameters fine-tuned using a stochastic fractal search-guided whale optimization algorithm. The model, comprising a convolutional neural network and long short-term memory layer, demonstrates superior performance on four diverse speech emotion datasets (IEMOCAP, Emo-DB, RAVDESS, and SAVEE). Experimental results showcase recognition respectively, highlighting the effectiveness and stability of the proposed methodology over existing state-of-the-art approaches.

OBJECTIVES

To rectify decryption limitations arising from the encryption scheme's pixel grouping method when encountering large integers surpassing the elliptic curve's prime parameter.

Addressing decryption inaccuracies caused by large integers surpassing the elliptic curve’s prime parameter in a pixel grouping-based encryption scheme.

INTRODUCTION

In the area of data transmission, safeguarding multimedia data from unauthorized access has become pivotal. Cryptographic methods serve this purpose but face challenges from cryptanalysis attacks, leading to continuous improvements. Chaotic system-based encryption gained traction due to its robustness and simplicity. Yet, methods like those using chaotic maps and information entropy encountered vulnerabilities, as highlighted by subsequent analyses. Various attempts emerged, such as combining Sine maps for encryption and utilizing multiple chaotic maps for bit plane manipulation. However, these methods faced vulnerabilities against chosen-plaintext and known-plaintext attacks. Advancements like hyper-chaotic systems seemed promising but succumbed to reversibility in permutation and stability in diffusion. Novel approaches, like employing DNA computing and hybrid models, showcased innovation but faced challenges in ensuring robust encryption.

LITERATURE SURVEY

1] Secure image transmission using chaotic-enhanced elliptic curve cryptography (Abdelfatah RI)

- In this paper, a novel scheme for secure and real time image transmission is introduced. The scheme uses a block-based elliptic curve (BBEC) public key encryption as a first stage of encryption so it solves the key distribution and management problem of symmetric key encryption in an efficient method.

- Then the security of first stage BBEC encryption is enhanced through a second stage of encryption which XOR the BBEC first stage output with a pseudo random sequence generated by a new designed multi chaotic pseudo random generator STH.

2] Cryptanalysis of a chaotic image encryption algorithm based on information entropy (Li C, Lin D, Feng B, L¨u J, Hao F)

Recently, a chaotic image encryption algorithm based on information entropy (IEAIE) was proposed. This paper scrutinizes the security properties of the algorithm and evaluates the validity of the used quantifiable security metrics

Even worse, each security metric is questionable, which undermines the security credibility of IEAIE. Hence, IEAIE can only serve as a counterexample for illustrating common pitfalls in designing secure communication method for image data.

3] Cryptanalysis and improvement in a plaintext-related image encryption scheme based on hyper chaos
(Liu L, Zhang Z, Chen R)

Recently, a plaintext-related image encryption scheme based on a hyper-chaotic system has been proposed by Li et al . In their encryption scheme, the permutation process is related to the sum of plaintext pixel values, and the diffusion process is related to the values of nine special positions in the permuted image. This can resist the existing chosen plaintext attack due to the ever-changing permutation matrix and diffusion matrix when the plaintext is changed. Here, we show that Li's encryption scheme contains two vulnerabilities.

4] Cryptanalysis and enhancement of an image encryption scheme based on bit-plane extraction and multiple chaotic maps (Liu Y, Qin Z, Wu J)

Recently, an image encryption scheme combining bit-plane extraction with multiple chaotic maps (IESBC) was proposed. The scheme extracts binary bit planes from the plain-image and performs bit-level permutation and confusion, which are controlled by a pseudo-random sequence and a random image generated by the Logistic map, respectively.

This paper analyzes the weak points of IESBC and proposes a known-plaintext attack and a chosen-plaintext attack on it.

5] Cryptanalysis of multimedia encryption using elliptic curve cryptography (Motilal KS, Singh LD, Tuithung T)

The encryption scheme proposed by Tawalbeh et al. It is based on elliptic curve cryptography (ECC). ECC depends on the difficulty to solve the elliptic curve discrete logarithmic problem. However we found that the order of Tawalbeh et al. elliptic curve is not large enough to protect from attacks like Baby Step, Pollard's Rho attack.

EXISTING SYSTEM

Now-a-days huge amount of multimedia images are floating in the network and to provide security to those images, encryption technologies will be using to encrypt those images and this encryption algorithms will secure images but it content will have high binary size which will take more network time to transfer.

DISADVANTAGES

1. Increased binary size affecting transmission.
2. Slower network transfer speed.
3. Potential network congestion concerns.
4. Demands higher bandwidth allocation.
5. Longer encryption process duration.
6. Challenges in real-time applications.
7. Increased data transmission costs.

PROPOSED SYSTEM

To reduce transmission time pixel grouping based chaotic elliptic curve cryptography was introduce which will read images and then group all pixels to form a lengthy encrypted integer which is lighter in network for transmission. Encrypted integer will be map to chaotic using XOR operation and while decrypting reverse Chaotic XOR will be applied and if lengthy integer size higher than prime value then decryption will not be accurate.

To avoid incorrect decryption we have added Inverse modulo technique which will reduce lengthy integer size to be in range of Prime value and then decryption will be accurate.

ADVANTAGES

1. Improved encryption scheme reliability.
2. Enhanced execution speed efficiency.
3. Enhanced decryption accuracy.
4. Optimized pixel grouping technique.
5. Increased encryption robustness.
6. Efficient data security measures.
7. Enhanced elliptic curve cryptography.

SYSTEM REQUIREMENTS AND SPECIFICATIONS

HARDWARE REQUIREMENTS:

RAM: 8 GB

Hard Disc or SSD: 500 GB

Processor: Intel 3rd generation or high

SOFTWARE REQUIREMENTS:

Operating system: Windows 10 /11

Software's : Python 3.7 or high version

IDLE : Jupiter Notebook

Thank You